

Trainings ▾

General Information

The primary functions of engine coolant include temperature control, freeze protection, liner pitting protection, corrosion protection, pH buffering, and scale and deposit protection. This procedures provides additional information on these functions.

Temperature Control

- Maintains engine temperature by circulating through cavities in the engine block, cylinder liners, turbocharger, exhaust gas recirculation (EGR) cooler, and other areas, and transferring heat to or from fluids and materials within the engine. Engine coolant is then pumped to the radiator or other heat exchangers to transfer waste heat to the environment. Improper coolant selection or management can lead to engine overheating.

Antifreeze and additives are necessary to provide other functions including:

Freeze Protection

- Preventing damage that could occur from freezing and expansion of aqueous heat transfer fluids.

Liner Pitting Protection

- Wet cylinder liner engines are subject to cylinder liner cavitation damage from the collapse of vapor bubbles in the liner cavity. Cavitation can cause severe pitting damage to cylinder liners that can result in coolant leaking directly into the combustion chamber.

Corrosion Protection

- Protects cooling system metals (iron, steel, aluminum, brass, lead solder) against corrosion that can lead to a variety of malfunctions within the cooling system and in the engine as a whole.

pH Buffering

- Resist changes in pH caused by exhaust gas leaks into coolant, coolant base degradation, general contamination, etc.

Scale and Deposit Prevention

- Scale and deposit buildup in the cooling system can prevent heat transfer and lead to localized overheating or hotspots.

